EXPERIMENT NO: 03

Loading Datasets (CSV, Excel, PDF) into SQLite using Python & Web Scraping

AIM

To load datasets from CSV, Excel, and PDF files into a SQLite database using Python, display the stored data, and perform simple web scraping to count HTML tables from a webpage.

PROCEDURE

- Import the required Python libraries (sqlite3, pandas, pdfplumber, requests, bs4).
- Connect to the SQLite database using Python's sqlite3 module.
- Load dataset files from CSV, Excel, and PDF formats.
- Store each dataset into a separate table in the SQLite database.
- Retrieve and display the stored table contents.
- Perform simple web scraping to count HTML tables from a target webpage.

PYTHON SOURCE CODE

```
import sqlite3
import pandas as pd
import pdfplumber
import requests
from bs4 import BeautifulSoup

# --- 1. Database Connection & Setup ---
conn = sqlite3.connect("hsd.db")
cursor = conn.cursor()

# --- 2. Load CSV Dataset ---
df_csv = pd.read_csv("hsd.csv")
df_csv.to_sql("hsd_csv", conn, if_exists="replace", index=False)
print("Loaded hsd_csv:", len(df_csv), "rows")

# --- 3. Load Excel Dataset ---
df_excel = pd.read_excel("hsd.xlsx")
df_excel.to_sql("hsd_excel", conn, if_exists="replace", index=False)
print("Loaded hsd_excel:", len(df_excel), "rows")

# --- 4. Load PDF Dataset ---
columns = [
    "age", "sex", "chest", "rest_bps", "cholestrol", "bloodreport",
    "resting_electrocardiographic_results", "max_heart_rate_achieved",
    "exercise_induced_angina", "oldpeak", "slope", "no_of_major_vessels", "thal", "class"
]
```

```
all_rows = []
with pdfplumber.open("data.pdf") as pdf:
    pages = pdf.pages
    for i in range(0, len(pages), 2):
        left = pages[i].extract_tables()[0]
        right = pages[i+1].extract_tables()[0]
        for lrow, rrow in zip(left[1:], right[1:]):
            all_rows.append(lrow + rrow)

df_pdf = pd.DataFrame(all_rows, columns=columns)
df_pdf.to_sql("hsd_pdf", conn, if_exists="replace", index=False)
print("Loaded hsd_pdf:", len(df_pdf), "rows")

conn.commit()

# --- 5. Retrieve & Display Database Contents ---
for table in ["hsd_csv", "hsd_excel", "hsd_pdf"]:
    print(f"\n--- {table} ---")
    print(pd.read_sql(f"SELECT * FROM {table} LIMIT 5;", conn))

conn.close()

# --- 6. Web Scraping to Count HTML Tables ---
url = "http://example.com"  # target URL
response = requests.get(url)
print("Response Status Code:", response.status_code)

soup = BeautifulSoup(response.text, "html.parser")
tables = soup.find_all("table")
print("Tables count:", len(tables))
```

OUTPUT DATA TABLES

Output 1: Sample Dataset Records

Age	Sex	Chest	BPs	Chol	BS	ECG	Max HR	Ex Angina	Oldpeak	Slope	Vessels	Thal	Class
70	1	4	130	322	0	2	109	0	2.4	2	3	3	present
67	0	3	115	564	0	2	160	0	1.6	2	0	7	absent
57	1	2	124	261	0	0	141	0	0.3	1	0	3	absent
64	1	4	128	263	0	0	105	1	0.2	2	1	7	absent
64	0	2	120	269	0	2	121	1	0.2	1	0	3	absent

Output 2: Scraping & Execution Summary

```
Loaded hsd_csv: 270 rows
```

```
Loaded hsd_excel: 270 rows
Loaded hsd_pdf: 270 rows
Response Status Code: 200
Tables count: 2
```

RESULT

Loaded datasets from CSV, Excel, and PDF files into SQLite database tables and displayed the data successfully.